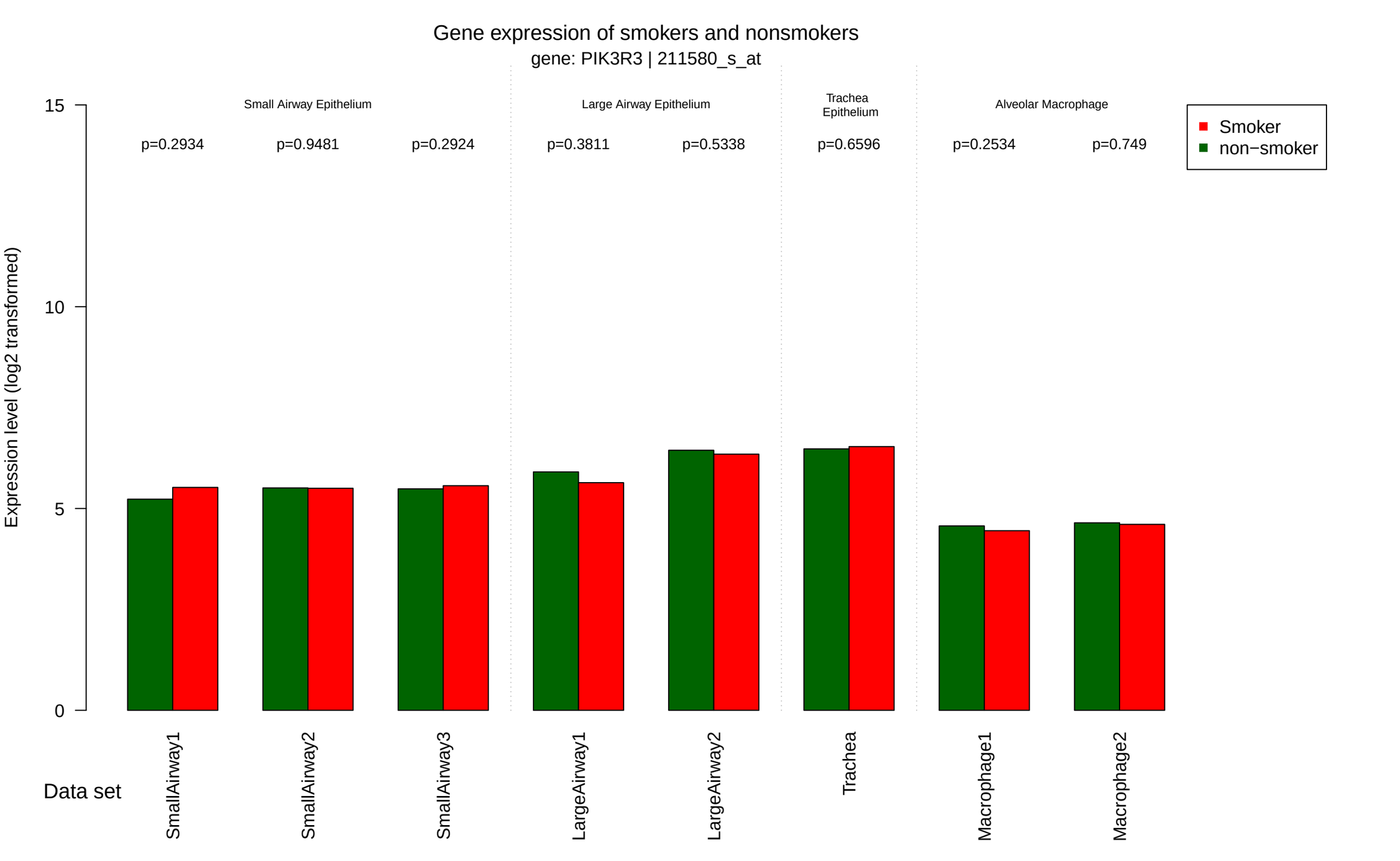
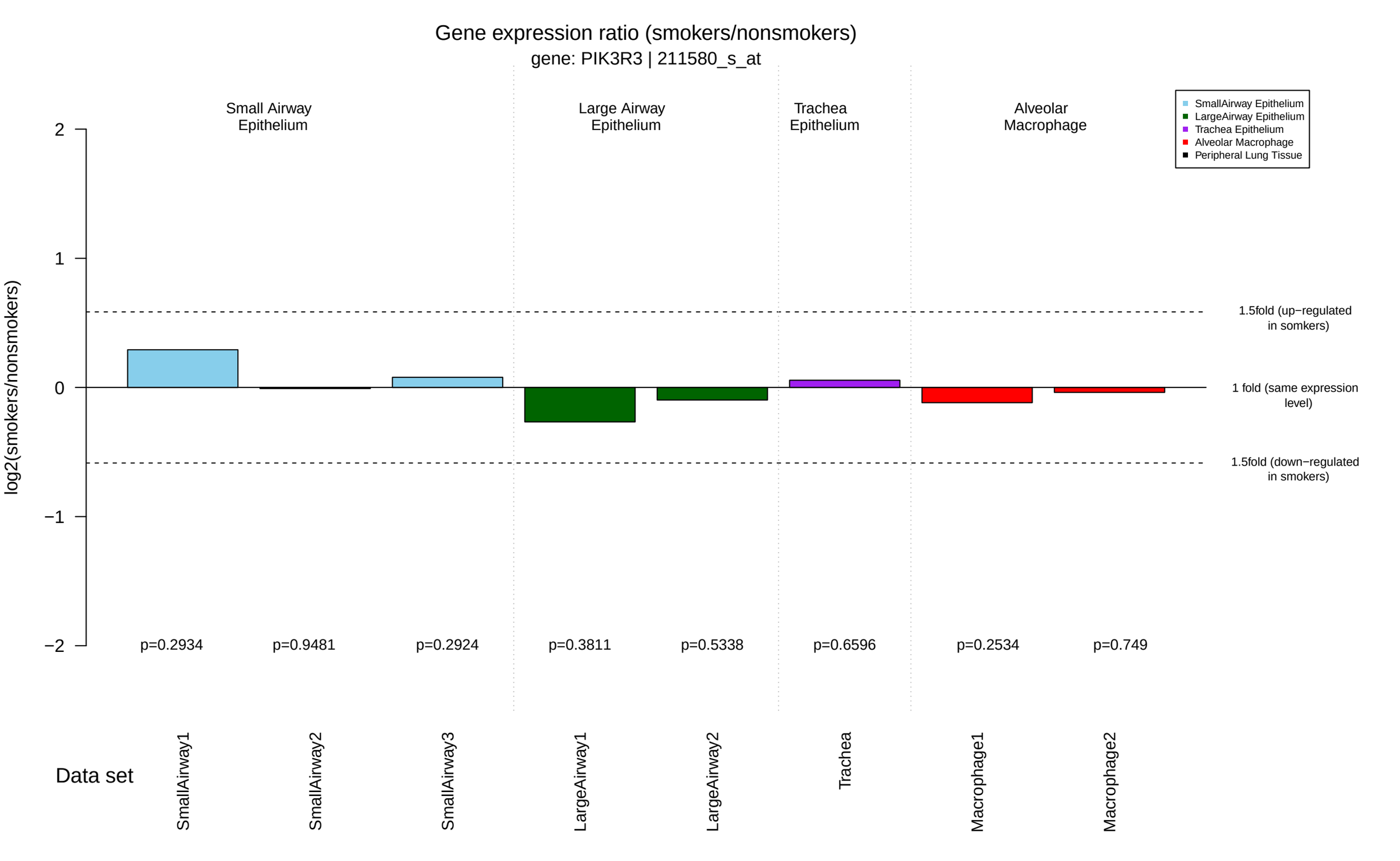
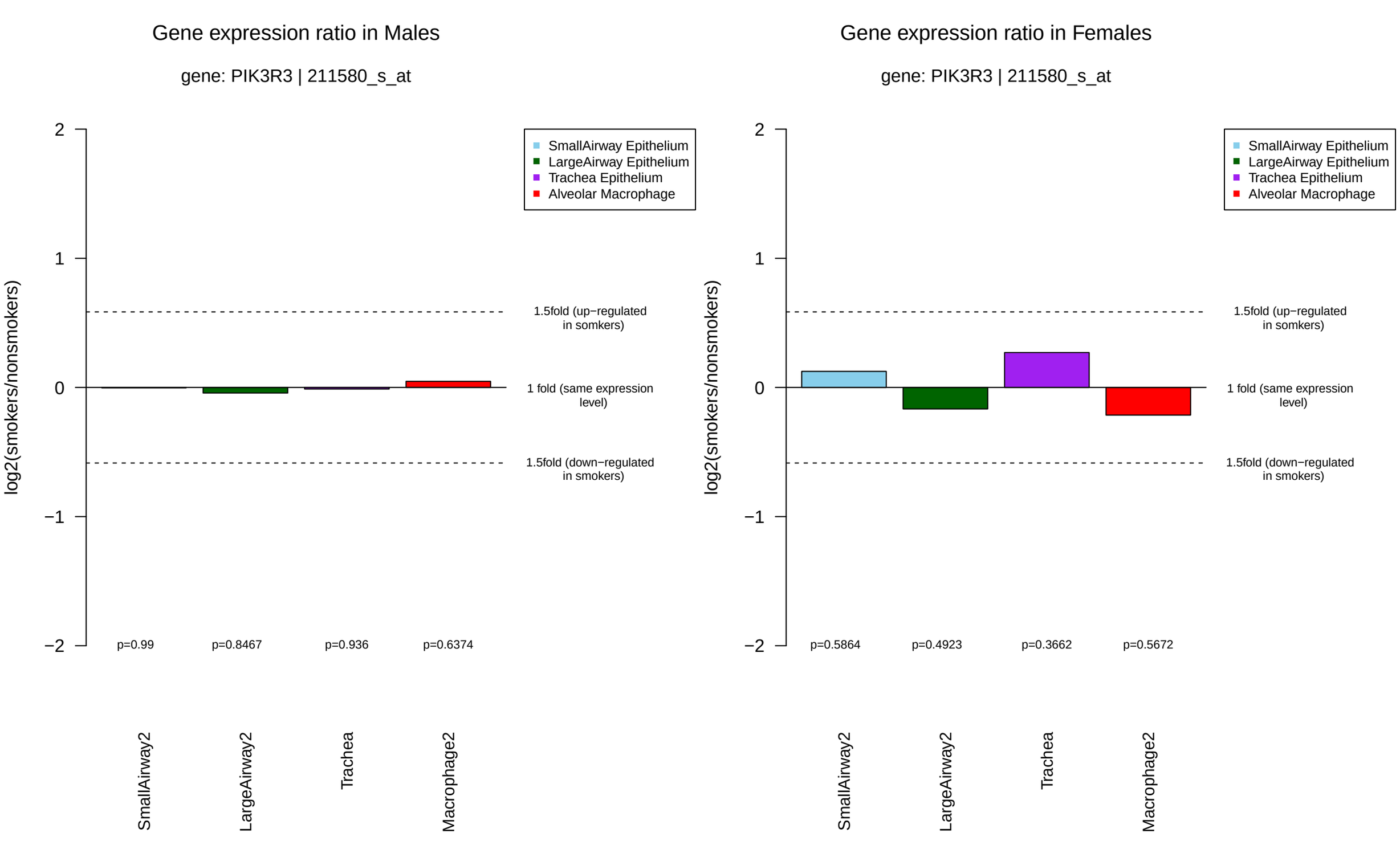
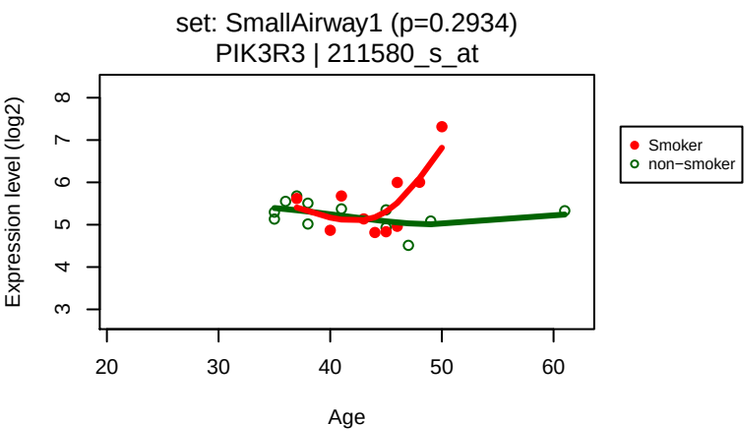
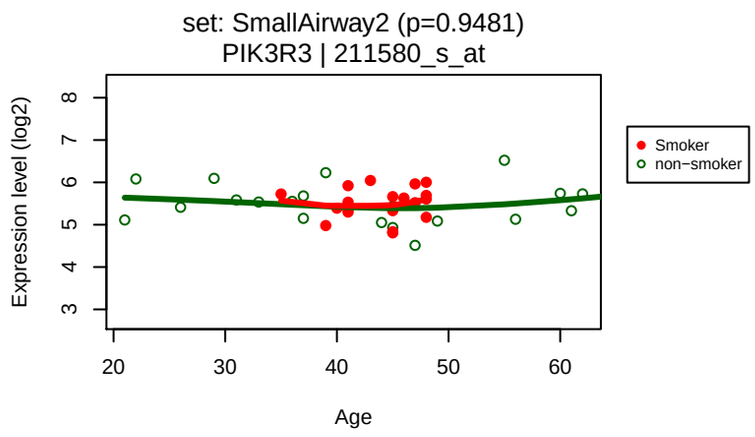
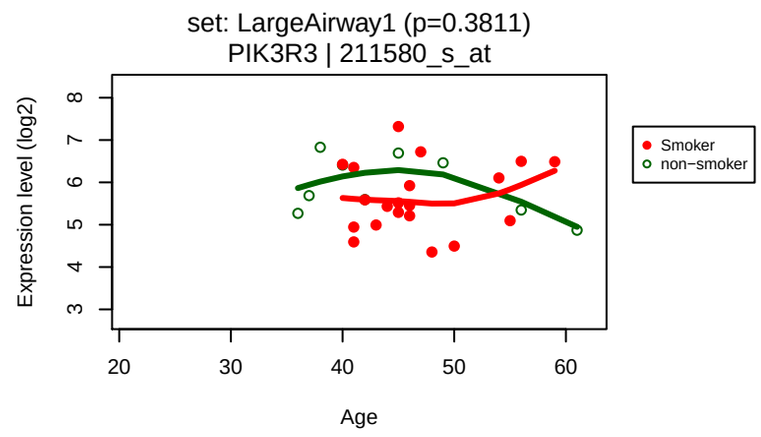

Gene expression of smokers and nonsmokers
gene: PIK3R3 | 211580_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2934
p=0.9481
p=0.2924
p=0.3811
p=0.5338
p=0.6596
p=0.2534
p=0.749
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

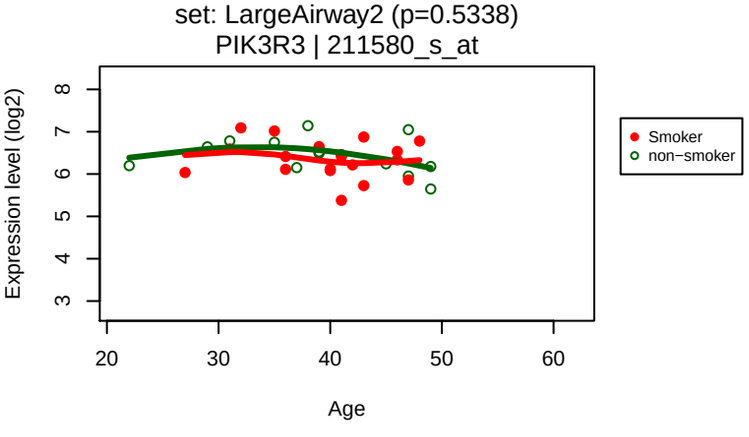

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R3 | 211580_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2934
p=0.9481
p=0.2924
p=0.3811
p=0.5338
p=0.6596
p=0.2534
p=0.749
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

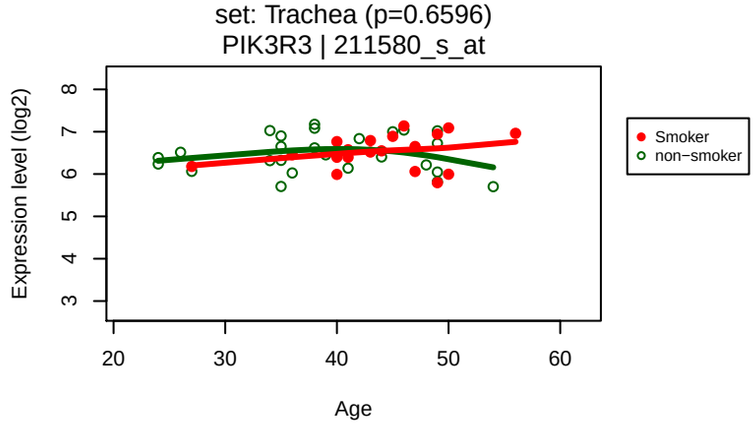

Gene expression ratio in Males
gene: PIK3R3 | 211580_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.99
p=0.8467
p=0.936
p=0.6374
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R3 | 211580_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5864
p=0.4923
p=0.3662
p=0.5672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

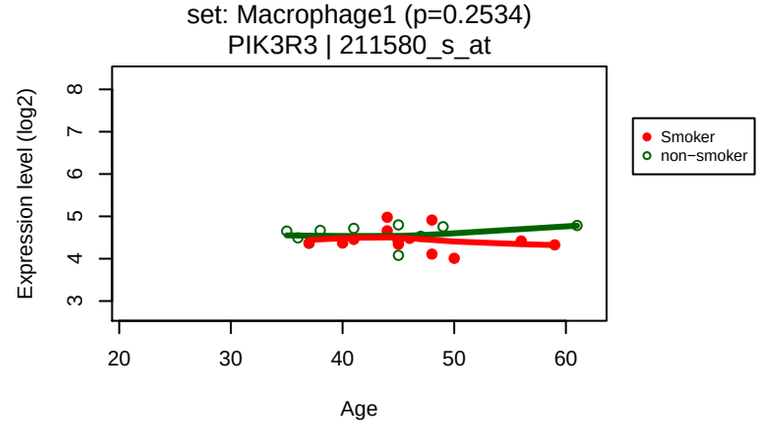

set: SmallAirway1 (p=0.2934)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker

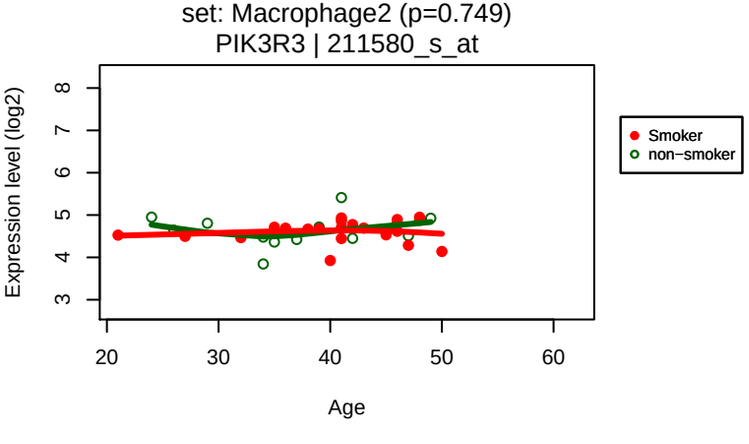

set: SmallAirway2 (p=0.9481)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker

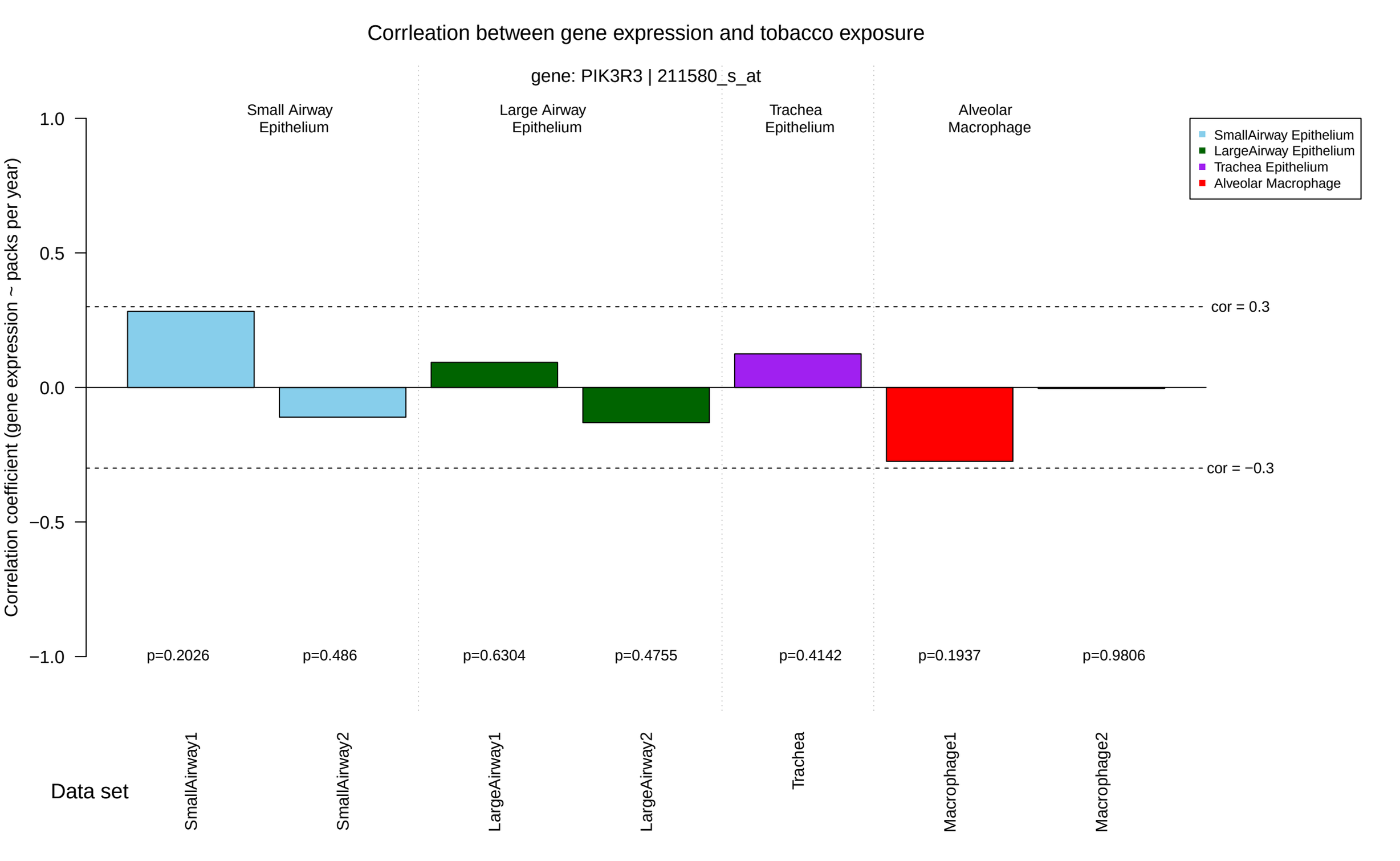

set: LargeAirway1 (p=0.3811)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5338)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6596)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2534)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.749)
PIK3R3 | 211580_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R3 | 211580_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2026
p=0.486
p=0.6304
p=0.4755
p=0.4142
p=0.1937
p=0.9806
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2